Lecture 11: Robust Optimization

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Review: Tractable Robust Counterparts

$$(\bar{a} + Pz)^T x \leq b, \quad \forall z \in Z.$$

Unc. set	Z	Robust Counterpart	Tractability
Box	$\|z\|_\infty \leq \rho$	$\bar{a}^T x + \rho \|P^T x\|_1 \leq b$	LO
Ellipsoidal	$\|z\|_2 \leq \rho$	$\bar{a}^T x + \rho \|P^T x\|_2 \leq b$	CQO
Polyhedral	$Dz \leq d$	$\begin{cases} \bar{a}^T x + d^T y \leq b \\ D^T y = P^T x \\ y \geq 0 \end{cases}$	LO
Budget	$\begin{cases} \|z\|_\infty \leq \rho \\ \|z\|_1 \leq \Gamma \end{cases}$	$\bar{a}^T x + \rho \|y\|_1 + \Gamma \|P^T x - y\|_\infty \leq b$	LO

RC for Box Uncertainty

$$(\bar{a} + Pz)^T x \leq b \qquad \forall z : \|z\|_\infty \leq \rho.$$

This is equivalent to:

$$\max_{z:\|z\|_\infty \leq \rho} (\bar{a} + Pz)^T x \leq b,$$

or

$$\bar{a}^T x + \max_{z:\|z\|_\infty \leq \rho} (P^T x)^T z \leq b,$$

or

$$\bar{a}^T x + \max_{z:|z_i| \leq \rho} \sum_i (P^T x)_i z_i \leq b,$$

or

$$\bar{a}^T x + \rho \sum_i |(P^T x)_i| \leq b,$$

or

$$\bar{a}^T x + \rho \|P^T x\|_1 \leq b.$$

RC for Ellipsoid Uncertainty

$$(\bar{a} + Pz)^T x \leq b \qquad \forall z : \|z\|_2 \leq \rho.$$

This is equivalent to:

$$\bar{a}^T x + \max_{z:\|z\|_2 \leq \rho} (P^T x)^T z \leq b.$$

Intermezzo: $\max \{c^T z : \|z\|_2 \leq \rho\}$ or $\max \{c^T z : z^T z \leq \rho\}$

Lagrange: $c = \lambda z$, and $\lambda = \|c\|_2 / \rho$.

Optimal objective value: $\frac{c^T c}{\lambda} = \rho \|c\|_2$.

Hence (RC) is:

$$\bar{a}^T x + \rho \|P^T x\|_2 \leq b.$$

RC for Polyhedron Uncertainty

$$(\bar{a} + Pz)^T x \le b \qquad \forall z \ : \ Dz \le d.$$

This is equivalent to:

$$\bar{a}^T x + \max_{z \ : \ Dz \le d} (P^T x)^T z \le b.$$

Note that by duality

$$\max\{(P^T x)^T z \ : \ Dz \le d\} = \min\{d^T y : D^T y = P^T x, \ y \ge 0\}.$$

Hence (7) is equivalent to

$$\bar{a}^T x + \min_y \{d^T y \ : \ D^T y = P^T x, \ y \ge 0\} \le b,$$

or

$$\bar{a}^T x + d^T y \le b, \quad D^T y = P^T x, \quad y \ge 0.$$

RC for Budget Uncertainty

Budget uncertainty set: $\{z \ : \ \|z\|_\infty \leq \rho, \ \|z\|_1 \leq \Gamma\}$,
which is equivalent to

$$\{z \ : \ z_i \leq \rho, \ -z_i \leq \rho \ \ \forall i, \ \sum_i w_i \leq \Gamma, \ w_i \geq z_i, \ w_i \geq -z_i \ \ \forall i\}.$$

- Less conservative than box uncertainty set (if Γ is small enough).
- Special case of polyhedral uncertainty set.

It can be shown that x satisfies

$$(\bar{a} + Pz)^T x \leq b \qquad \forall z : \|z\|_\infty \leq \rho, \ \|z\|_1 \leq \Gamma.$$

if and only if there exists a y such that (x, y) satisfies

$$\bar{a}^T x + \rho\|y\|_1 + \Gamma\|P^T x - y\|_\infty \leq b.$$

Another interpretation of Budget Uncertainty

(Bertsimas and Sim (2004))

Wlog let us assume that $\rho = 1$, and moreover that Γ is integer.

We consider the following **cardinality constrained set**:

$$Z = \{z \mid \|z\|_\infty \leq 1, \ |N(z)| \leq \Gamma\},$$

where $N(z) = \{i \mid z_i \neq 0\}$.

It allows at most Γ coefficients to deviate from the nominal value $z = 0$.

Worst-case scenario for **budget uncertainty set** is also such that at most Γ coefficients deviate from $z = 0$.

Hence, the **budget uncertainty** and the **cardinality constrained set** lead to exactly the same robust counterpart.

Goal: General Convex Constraints

$$(\bar{a} + Pz)^T x \leq b, \quad \forall z \in Z.$$

Unc. set	Z	Robust Counterpart	Tractability
Box	$\|z\|_\infty \leq \rho$	$\bar{a}^T x + \rho \|P^T x\|_1 \leq b$	LO
Ellipsoidal	$\|z\|_2 \leq \rho$	$\bar{a}^T x + \rho \|P^T x\|_2 \leq b$	CQO
Polyhedral	$Dz \leq d$	$\begin{cases} \bar{a}^T x + d^T y \leq b \\ D^T y = P^T x \\ y \geq 0 \end{cases}$	LO
Budget	$\begin{cases} \|z\|_\infty \leq \rho \\ \|z\|_1 \leq \Gamma \end{cases}$	$\bar{a}^T x + \rho \|y\|_1 + \Gamma \|P^T x - y\|_\infty \leq b$	LO
Convex constr.	$h_k(z) \leq 0$ $k = 1, \ldots, K$	$\begin{cases} \bar{a}^T x + \sum_k u_k h_k^* \left(\frac{w^k}{u_k} \right) \leq b \\ \sum_k w^k = P^T x \\ u \geq 0 \end{cases}$	Convex

Preliminary: Support Function

Let $f : \mathrm{R}^n \to \mathrm{R}$ be convex with $\mathrm{dom}(f) = \{x \mid f(x) < \infty\}$.

The convex conjugate of f is defined as

$$f^*(y) = \sup_{x \in \mathrm{dom}(f)} \{y^T x - f(x)\}.$$

The indicator function on the set S is defined as:

$$\delta(x|S) = \begin{cases} 0 & \text{if} \quad x \in S \\ \infty & \text{otherwise.} \end{cases}$$

Then the conjugate function of $\delta(x|S)$,

$$\delta^*(y|S) = \sup_x \{y^T x - \delta(x|S)\} = \sup_{x \in S} y^T x,$$

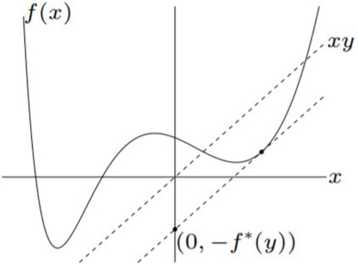
is the so-called support function of the set S.

Graphical explanation of conjugate

The conjugate function

the **conjugate** of a function f is

$$f^*(y) = \sup_{x \in \mathbf{dom}\, f} \left(y^T x - f(x)\right)$$



Conjugate example 1

$$f^*(y) = \sup_{x \in \mathrm{dom}(f)} \{y^T x - f(x)\}.$$

Example: $f(x) = x^2$.

We have:

$$f^*(y) = \sup_x \{yx - x^2\}.$$

For maximum it should hold: $y - 2x = 0$, hence $x = y/2$.

So:

$$f^*(y) = y.y/2 - (y/2)^2 = y^2/4.$$

Conjugate example 2

$$f^*(y) = \sup_{x \in \mathrm{dom}(f)} \{y^T x - f(x)\}.$$

Example: $f(x) = -\ln(x)$.

We have $\mathrm{dom}(f)$ is $x > 0$, and:

$$f^*(y) = \sup_{x>0} \{yx + \ln(x)\}.$$

For maximum it should hold: $y + 1/x = 0$, hence $x = -1/y$, for $y < 0$.

So:

$$f^*(y) = \begin{cases} -1 - \ln(-y) & y < 0 \\ \infty & y \geq 0. \end{cases}$$

Conjugate example 3

$$f^*(y) = \sup_{x \in \mathrm{dom}(f)} \{y^T x - f(x)\}.$$

Example: $f(x) = x^T Q x + b^T x + c$, $x \in \mathrm{R}^n$, Q positive definite.

We have:

$$f^*(y) = \sup_x \{y^T x - x^T Q x - b^T x - c\}.$$

For maximum it should hold: $y - 2Qx - b = 0$, hence $x = Q^{-1}(y - b)/2$.

So:

$$\begin{aligned} f^*(y) &= (y-b)^T Q^{-1}(y-b)/2 - (y-b)^T Q^{-1} Q Q^{-1}(y-b)/4 - c \\ &= (y-b)^T Q^{-1}(y-b)/4 - c. \end{aligned}$$

Conjugate example 4

$$f^*(y) = \sup_{x \in \mathrm{dom}(f)} \{y^T x - f(x)\}.$$

Example: $f(x) = a^T x - b$, $x \in \mathrm{R}^n$.

We have:

$$f^*(y) = \sup_x \{y^T x - a^T x + b\} = \sup_x \{(y-a)^T x + b\}.$$

So:

$$f^*(y) = \begin{cases} b & y = a \\ \infty & y \neq a. \end{cases}$$

Conjugate property 1

For $a > 0$

$$(af)^*(y) = af^*\left(\frac{y}{a}\right),$$

and for $\tilde{f}(x) = f(ax)$ we have

$$\tilde{f}^*(y) = f^*\left(\frac{y}{a}\right)$$

and for $\tilde{f}(x) = f(x - a)$ we have

$$\tilde{f}^*(y) = f^*(y) + a^T y.$$

Conjugate property 2

Lemma

Assume that f_i, $i = 1, ..., m$, *are convex, and the intersection of the relative interiors of the domains of* f_i, $i = 1, ..., m$, *is nonempty, i.e.,* $\cap_{i=1}^m ri(dom\, f_i) \neq \emptyset$. *Then*

$$\left(\sum_{i=1}^m f_i\right)^*(s) = \inf_{\{v^i\}_{i=1}^m} \left\{\sum_{i=1}^m f_i^*(v^i) \mid \sum_{i=1}^m v^i = s\right\},$$

and the inf *is attained for some* v_i, $i = 1, ..., m$. □

Preliminary: Support Function

Let $f : \mathrm{R}^n \to \mathrm{R}$ be convex with $\mathrm{dom}(f) = \{x \mid f(x) < \infty\}$.

The convex conjugate of f is defined as

$$f^*(y) = \sup_{x \in \mathrm{dom}(f)} \{y^T x - f(x)\}.$$

The indicator function on the set S is defined as:

$$\delta(x|S) = \begin{cases} 0 & \text{if} \quad x \in S \\ \infty & \text{otherwise.} \end{cases}$$

Then the conjugate function of $\delta(x|S)$,

$$\delta^*(y|S) = \sup_x \{y^T x - \delta(x|S)\} = \sup_{x \in S} y^T x,$$

is the so-called support function of the set S.

Basic robust counterpart

Robust counterpart for the robust linear constraint

$$(\bar{a} + Pz)^T x \leq b \quad \forall z \in Z,$$

can be written as: $\bar{a}^T x + \left(P^T x\right)^T z \leq b \quad \forall z \in Z$,

and as: $\bar{a}^T x + \max_{z \in Z} \left(P^T x\right)^T z \leq b$.

Using the definition of support function, this constraint can be written as

$$(RC) \qquad \bar{a}^T x + \delta^*(P^T x | Z) \leq b.$$

The safety term is the support function of Z evaluated at $P^T x$.

Closed-form expression for support function not needed!

Computing δ^*: 2-norm

$$(RC) \qquad \bar{a}^T x + \delta^*(P^T x | Z) \leq b.$$

Let $Z = \{z \mid \|z\|_2 \leq \rho\}$.

For the support function of Z we have:

$$\delta^*(y|Z) = \sup_{x \,:\, \|x\|_2 \leq \rho} y^T x = \rho \|y\|_2.$$

Hence, RC is equivalent to

$$\bar{a}^T x + \rho \|P^T x\|_2 \leq b.$$

Computing δ^*: p-norm

Let $Z = \{z \mid \|z\|_p \le \rho\}$, where $\|.\|_p$ is the p-norm.

For the support function of Z we have:

$$\delta^*(y|Z) = \sup_{x\,:\,\|x\|_p \le \rho} y^T x = \rho\|y\|_p^* = \rho\|y\|_q,$$

where $\|.\|_p^*$ is the dual norm of the p-norm, i.e., $1/p + 1/q = 1$.

p	q
2	2
1	∞
∞	1

Hence, $\bar{a}^T x + \delta^*(P^T x|Z) \le b$ is equivalent to: $\bar{a}^T x + \rho\|P^T x\|_q \le b$.

Computing δ^*: polyhedron

Let $Z = \{z \mid Dz \leq d\}$.

For the support function of Z we have:

$$\delta^*(y|Z) = \max_z\{y^T z \mid Dz \leq d\} = \min_u\{d^T u \mid D^T u = y,\ u \geq 0\},$$

where the last equality follows from LO duality.

Since RC is a "$\leq$" inequality, we may omit the min.

Hence, $\bar{a}^T x + \delta^*(P^T x|Z) \leq b$ is equivalent to

$$\begin{cases} \bar{a}^T x + d^T u \leq b \\ D^T u = P^T x \\ u \geq 0. \end{cases}$$

Computing δ^*: convex constraints

$$(RC) \qquad \bar{a}^T x + \delta^*(P^T x|Z) \leq b.$$

Lemma

Let $Z = \{z \mid h_k(z) \leq 0,\ k = 1, ..., K\}$, where $h_k(.)$ is convex. Moreover, we assume that $\cap_{k=1}^{K} ri(dom\, h_k) \neq \emptyset$. Then

$$\delta^*(y|Z) = \min_{u \geq 0} \left\{ \sum_{k=1}^{K} u_k h_k^* \left(\frac{v^k}{u_k} \right) \mid \sum_{k=1}^{K} v^k = y \right\}. \qquad (2)$$

Then (RC) becomes:

$$\begin{cases} \bar{a}^T x + \sum_{k=1}^{K} u_k h_k^* \left(\frac{v^k}{u_k} \right) \leq b \\ \sum_{k=1}^{K} v^k = P^T x \\ u \geq 0. \end{cases}$$

Proof

It can easily be verified that

$$
\begin{aligned}
\delta^*(y|Z) &= \sup_z \{y^T z \mid h_k(z) \le 0,\ k = 1, \ldots, K\} \\
&= \inf_{u \ge 0} \sup_z \left\{ y^T z - \sum_{k=1}^{K} u_k h_k(z) \right\} \\
&= \inf_{u \ge 0} \left(\sum_{k=1}^{K} u_k h_k \right)^* (y) \\
&= \inf_{u \ge 0, \{v_k\}} \left\{ \sum_{k=1}^{K} u_k h_k^* \left(\frac{v^k}{u_k} \right) \;\middle|\; \sum_{k=1}^{K} v^k = y \right\},
\end{aligned}
$$

where the last equality follows by Lemma 1 and property (1).

Computing δ^*: intersection

$$(RC) \qquad \bar{a}^T x + \delta^*(P^T x | Z) \leq b.$$

Lemma

Let $Z_1, ..., Z_k$ be closed convex sets, such that $\bigcap_i ri(Z_i) \neq \emptyset$, and let $Z = \cap_{i=1}^k Z_i$. Then

$$\delta^*(y|Z) = \min_{y^1, \cdots, y^k} \left\{ \sum_{i=1}^k \delta^*(y^i | Z_i) \mid \sum_{i=1}^k y^i = y \right\}.$$

and (RC) becomes

$$\begin{cases} \bar{a}^T x + \sum_{i=1}^k \delta^*(y^i | Z_i) \leq 0 \\ \sum_{i=1}^k y^i = P^T x \end{cases}$$

Proof: Use $\delta^*(y|Z) = (\delta(y|Z_1) + \ldots + \delta(y|Z_k))^*$.

Computing δ^*: Budget constraint

Budget uncertainty set: $Z = \{z \ : \ \|z\|_\infty \le \rho, \ \|z\|_1 \le \Gamma\}$.

Define: $Z_1 = \{z \ : \ \|z\|_1 \le \Gamma\}$ and $Z_\infty = \{z \ : \ \|z\|_\infty \le \rho\}$.

Then: $Z = Z_1 \cap Z_\infty$ and

$$\begin{aligned}\delta^*(y|Z) &= \min_{w^1, w^\infty} \{\delta^*(w^1|Z_1) + \delta^*(w^\infty|Z_\infty) \mid w^1 + w^\infty = y\} \\ &= \min_{w^1, w^\infty} \{\Gamma\|w^1\|_\infty + \rho\|w^\infty\|_1 \mid w^1 + w^\infty = y\}.\end{aligned}$$

Hence, $\bar{a}^T x + \delta^*(P^T x|Z) \le b$ is equivalent to

$$\begin{cases} \bar{a}^T x + \Gamma\|w^1\|_\infty + \rho\|w^\infty\|_1 \le b \\ w^1 + w^\infty = P^T x. \end{cases}$$

or $\bar{a}^T x + \Gamma\|w^1\|_\infty + \rho\|P^T x - w^1\|_1 \le b$.

Computing δ^*:Ball-Box

Let $Z_2 = \{z \mid \|z\|_2 \leq \rho_2\}$ and $Z_\infty = \{z \mid \|z\|_\infty \leq \rho_\infty\}$.

Let $Z = Z_2 \cap Z_\infty$.

Then

$$\begin{aligned}
\delta^*(y|Z) &= \min_{w^2, w^\infty} \{\delta^*(w^2|Z_2) + \delta^*(w^\infty|Z_\infty) \mid w^2 + w^\infty = y\} \\
&= \min_{w^2, w^\infty} \{\rho_2\|w^2\|_2 + \rho_\infty\|w^\infty\|_1 \mid w^2 + w^\infty = y\}.
\end{aligned}$$

Hence, $\bar{a}^T x + \delta^*(P^T x|Z) \leq b$ is equivalent to

$$\begin{cases}
\bar{a}^T x + \rho_2\|w^2\|_2 + \rho_\infty\|w^\infty\|_1 \leq b \\
w^2 + w^\infty = P^T x.
\end{cases}$$

Example: single-period portfolio selection

- There are 200 assets.
- Asset # 200 (“money in the bank”) has yearly return $r_{200} = 1.05$ and zero variability.
- The yearly returns r_l, $l = 1, \ldots, 199$, of the remaining assets are independent random variables taking values in the segments $[\mu_l - \sigma_l, \mu_l + \sigma_l]$ with expected values μ_l.
- Here

$$\mu_l = 1.05 + 0.3\frac{(200-l)}{199}, \quad \sigma_l = 0.05 + 0.6\frac{(200-l)}{199}, \quad l = 1, \ldots, 199.$$

- Goal: distribute \$1 between the assets in order to maximize the return of the resulting portfolio, the required risk level is 0.5%.

Example: single-period portfolio selection

We want to solve the uncertain LO problem

$$\max_{x,t}\{t \;:\; \sum_{l=1}^{199} r_l x_l + r_{200}x_{200} \geq t,\; \sum_{l=1}^{200} x_l = 1,\; x \geq 0\},$$

where x_l is the capital to be invested into asset l.

The uncertain data are the returns $r_l = \mu_l + \sigma_l z_l$, $l = 1, ..., 199$; where z_l, $l = 1, ..., 199$, are independent random pertubations with zero mean varying in the segments $[-1, 1]$.

The RC is:

$$\begin{cases} \max\limits_{x,t} \quad t \\ \sum_{l=1}^{199}(\mu_l + \sigma_l z_l)x_l + 1.05x_{200} \geq t, \quad \forall z \in Z \\ \sum_{l=1}^{200} x_l = 1 \\ x \geq 0. \end{cases}$$

Example: single-period portfolio selection

We consider 3 uncertainty sets:

1. Box: $Z = \{z : \|z\|_\infty \leq 1\}$

   Uses the only fact that uncertainties vary $[-1, 1]$.

2. Ball-Box: $Z = \{z : \|z\|_\infty \leq 1\ ,\ \|z\|_2 \leq 3.255\}$,

   with the safety parameter $\Omega = \sqrt{2\ln(1/\epsilon)} = 3.255$

   which ensures that the optimal solution of the RC satisfies the constraint with probability at least $1 - \epsilon = 0.995$.

3. Budget: $Z = \{z : \|z\|_\infty \leq 1\ ,\ \|z\|_1 \leq 45.921\}$,

   with the safety parameter $\Gamma = \sqrt{2\ln(1/\epsilon)}\sqrt{L} = 45.921$. $(L = 199)$

   which ensures that the optimal solution of the RC satisfies the constraint with probability at least $1 - \epsilon = 0.995$.

Box

The associated RC is the LO problem:

$$\begin{cases} \max\limits_{x,t} t \\ \sum_{l=1}^{199}(\mu_l - \sigma_l)x_l + 1.05x_{200} \geq t \\ \sum_{l=1}^{200} x_l = 1 \\ x \geq 0. \end{cases}$$

Worst possible values $r_l = \mu_l - \sigma_l, \quad l = 1, ..., 199$, of the uncertain returns.

These values are less than the guaranteed return for money.
Robust optimal solution: keep initial capital in the bank.

Guaranteed yearly return 1.05.

Ball-box

The associated RC is the conic quadratic problem

$$\begin{cases} \max\limits_{x,t,v,w} \; t \\ \sum_{l=1}^{199} \mu_l x_l + 1.05 x_{200} - \sum_{l=1}^{199} |v_l| - 3.255\sqrt{\sum_{l=1}^{199} w_l^2} \geq t \\ v_l + w_l = \sigma_l x_l, \quad l = 1, ..., 199 \\ \sum_{l=1}^{200} x_l = 1 \\ x \geq 0. \end{cases}$$

The robust optimal value is 1.12, meaning 12.0% profit

with risk as low as $\epsilon = 0.5\%$.

Budget

The associated RC is the conic quadratic problem

$$\begin{cases} \max\limits_{x,t,v,w} \; t \\ \sum_{l=1}^{199} \mu_l x_l + 1.05 x_{200} - \sum_{l=1}^{199} |v_l| - 45.921 \max_{l=1,\ldots,199} |w_l| \geq t \\ v_l + w_l = \sigma_l x_l, \quad l = 1, \ldots, 199 \\ \sum_{l=1}^{200} x_l = 1 \\ x \geq 0. \end{cases}$$

The robust optimal value is 1.10, meaning 10.0% profit

with risk as low as $\epsilon = 0.5\%$.